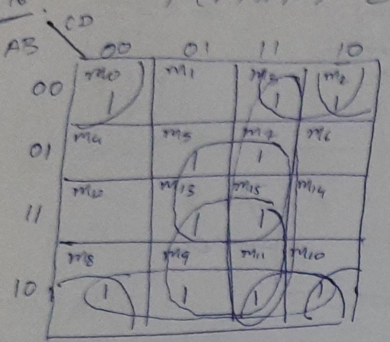
# 2016

12/8/16 $F(A,B,C,D) = \sum(0,2,3,5,7,8,9,10,11,13,15)$




One K-map, multiple function.

$F_1 = BD + B'D' + \overset{AB'}{(8,9,11,10)} + \overset{B'D}{(3,2,11,10)}$

$F_2 = BD + B'D' + \overset{AB'}{(8\ 9\ 11\ 10)} + \overset{CD}{(3\ 7\ 15\ 11)}$

$F_3 = BD + B'D' + \overset{AC}{(13\ 15\ 9\ 11)} + \overset{B'D}{(3\ 2\ 11\ 10)}$

$F_4 = BD + B'D' + \overset{AC}{(13\ 15\ 9\ 11)} + \overset{CD}{(3\ 7\ 15\ 11)}$

## Adder:

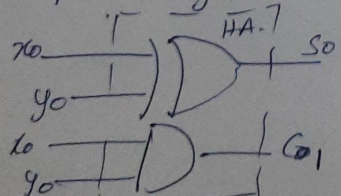
1 bit half adder:

| | | | | |
|---|---|---|---|---|
| $x_0$ | 0 | 0 | 1 | 1 |
| $y_0$ | 1 | 0 | 0 | 1 |
| | 1 | 0 | 1 | 10 |
| | $S_0$ $C_0$ | | | $C_0$ $S_0$ |

| $x_0$ | $y_0$ | $S_0$ | $C_{01}$ |
|---|---|---|---|
| 0 | 0 | 0 | 0 |
| 0 | 1 | 1 | 0 |
| 1 | 0 | 1 | 0 |
| 1 | 1 | 0 | 1 |

$S_0 = x_0 y_0' + x_0' y_0 = x_0 \oplus y_0$

$C_0 = x_0 y_0$




1 bit full adder

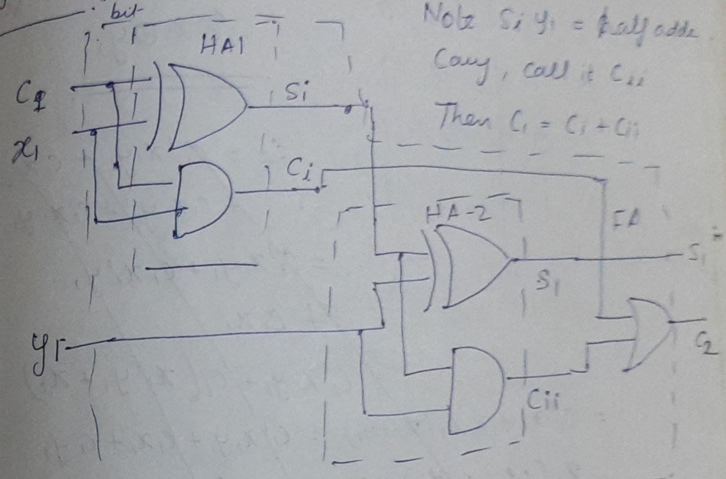
| | | |
|---|---|---|
| $C_j$ | $q_i$ | $C_{01}$ |
| $x_j$ | $x_i$ | $x_1$ |
| $C_{j+1}$ $S_j$ int | $y_i$ | $C_i$ $S_i$ |
| $y_j$ | | $y_1$ |
| $C_{j+1}$ $S_j \to j^{th}$ bit | | $C_2$ $S_1$ |

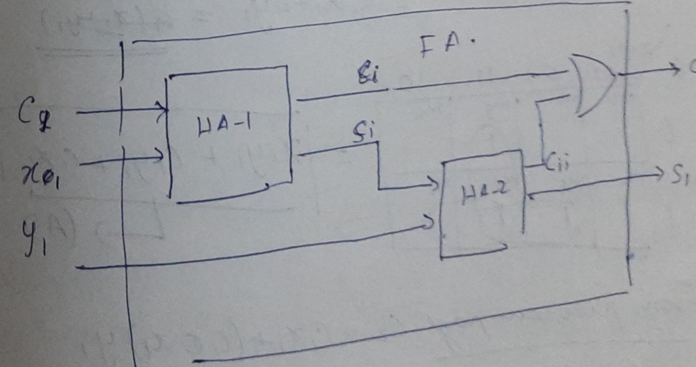
$\boxed{S_1 = S_i \oplus y_1 = C_0 \oplus x_1 \oplus y_1}$

$C_2 = C_i + S_i y_1$
$= C_i + (C_0 \oplus x_1) y_1$
$= C_0 x_1 + (C_0 \oplus x_1) y_1$

Note $S_i y_1$ = half adder carry, call it $C_{ii}$

Then $C_1 = C_i + C_{ii}$



| $C_1$ | $x_1$ | $y_1$ | $S_1$ | $C_2$ |
|---|---|---|---|---|
| 0 | 0 | 0 | 0 | 0 |
| 0 | 0 | 1 | 1 | 0 |
| 0 | 1 | 0 | 1 | 0 |
| 0 | 1 | 1 | 0 | 1 |
| 1 | 0 | 0 | 1 | 0 |
| 1 | 0 | 1 | 0 | 1 |
| 1 | 1 | 0 | 0 | 1 |
| 1 | 1 | 1 | 1 | 1 |

$S_1 = C_1'x_1'y_1 + C_1'x_1y_1' + C_1x_1'y_1' + C_1x_1y_1$

$= C_1'(x_1 \oplus y_1) + C_1(x_1 \oplus y_1)'$

X-NOR

$= C_1 \oplus x_1 \oplus y_1$

$C_2 = C_1'x_1y_1 + C_1x_1'y_1 + C_1x_1y_1' + C_1x_1y_1$

$= C_1'x_1y_1 + C_1x_1'y_1 + C_1x_1$

$= C_1'x_1y_1 + C_1(x_1'y_1 + x_1)$

$= C_1'x_1y_1 + C_1(x_1 + y_1) = C_1'x_1y_1 + C_1x_1 + C_1y_1$

$= x_1(C_1' + C_1y_1) + C_1y_1 = x_1(C_1 + y_1) + C_1y_1$

$= C_1x_1 + C_1y_1 = \cancel{C_1(x_1+y_1)}$

| $C_1$ \ $x_1y_1$ | 00 | 01 | 11 | 10 |
|---|---|---|---|---|
| 0 | $m_0$ | $m_1$ | $m_3$ 1 | $m_2$ |
| 1 | $m_4$ | $m_5$ 1 | $m_7$ 1 | $m_6$ 1 |

$= \boxed{x_1y_1 + C_1y_1 + C_1x_1} \rightarrow (A)$

From previous page: $C_2 = C_1x_1 + (C_1 \oplus x_1)y_1$

$= C_1x_1 + C_1x_1'y_1 + C_1'x_1y_1$

$= \cancel{C_1x_1+}\ C_1(x_1 + x_1'y_1) + C_1'x_1y_1$

$= C_1x_1 + C_1y_1 + C_1'x_1y_1$

$= C_1x_1 + y_1(C_1 + C_1'x_1) = C_1x_1 + y_1(C_1 + x_1)$

$= \boxed{C_1x_1 + C_1y_1 + x_1y_1} \rightarrow (B)$

$\boxed{(A) = (B)}$ !!

23-8-16 **Look Ahead Carry generator:**

Define 2 variable $P_i = A_i \oplus B_i$, $G_i = A_iB_i$

$C_{i+1} = G_i + P_iC_i$ (Note this is same as the expression $C_{i+1} = C_i + S_iB_i(y_i)$ derived before)

$G_i$ = carry generate.

$P_i$ = carry propagate (propagates carry from previous stage).

$P_0 = A_0 \oplus B_0 \rightarrow (1)$. $G_0 = A_0B_0 \rightarrow (2)$

$C_1 = G_0 + P_0C_0 \rightarrow (3)$.

$C_2 = G_1 + P_1C_1 = G_1 + P_1(G_0 + P_0C_0)$

$= G_1 + P_1G_0 + P_1P_0C_0 \rightarrow (4)$.

$C_3 = G_2 + P_2C_2 = G_2 + P_2(G_1 + P_1C_1)$

$= G_2 + P_2(G_1 + P_1(G_0 + P_0C_0))$

$= G_2 + P_2G_1 + P_2P_1G_0 + P_2P_1P_0C_0 \rightarrow (5)$.

$C_4 = G_3 + P_3C_3 = G_3 + P_3(G_2 + P_2C_2)$

$= G_3 + P_3(G_2 + P_2(G_1 + P_1C_1)) = G_3 + P_3(G_2 + P_2(G_1 + P_1(G_0 + P_0C_0)))$

$= G_3 + P_3G_2 + P_3P_2G_1 + P_3P_2P_1G_0 + P_3P_2P_1P_0C_0 \rightarrow (6)$